

PRAKHAR JAISWAL

prakhrajaiswal221@gmail.com | +91 70046 02209 | [LinkedIn](#) | [GitHub](#)

SUMMARY

Computer Science undergraduate with hands-on experience in AI/ML systems, full-stack development, and cloud deployment. Worked on medical image analysis and document retrieval applications using PyTorch and modern ML pipelines. Experienced in AWS-based deployments, databases, and collaborative software development, with a strong interest in automation and scalable engineering solutions.

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

2022 – 2026

CGPA: 8.20

SKILLS

Languages: Java, JavaScript, Python

Databases: MySQL, PostgreSQL, MongoDB

AI & ML: PyTorch, RAG, GenAI, LLMs, Computer Vision

Frameworks: React, Express, Mongoose

Tools: GitHub, Postman, Amazon Web Service, Docker, SSMS, Visual Studio Code

PROFESSIONAL EXPERIENCE

Tata Steel — Vocational Trainee (12 Weeks) [\[Link\]](#)

Oct 2024 – Dec 2024

Jamshedpur, Jharkhand

Contributed to the development of a hospital management system as part of an industrial training project. Collaborated effectively within a 4-member development team to meet functional and technical requirements. Gained practical exposure to the software development lifecycle and industry-grade deployment workflows. Assisted in application development tasks based on technical and functional requirements.

PROJECTS

Brain Tumor Segmentation - AI [\[GitHub\]](#) — Tech Stack: nnU-Net 2025, PyTorch, AWS Fargate

March 2025 – May 2025

Developed and tested an end-to-end medical image analysis application, optimizing performance and applying analytical debugging for reliable results.

Optimized inference to **5.8 seconds per scan** using GPU acceleration and efficient preprocessing.

End-to-end AI/ML pipeline: data prep to optimized, scalable AWS Fargate production deployment.

Scholar Assist AI [\[GitHub\]](#) — Tech Stack: Python, OpenAI Embeddings, LiteLLM

Dec 2024 – Feb 2025

Built a document retrieval application with modular, testable components, strengthening application logic and efficient data handling.

Achieved low-latency retrieval with agentic workflows and re-ranking across repositories of **100+ papers**.

Enabled precise retrieval using configurable embeddings and evidence sources, with responses generated in **under 5 seconds**, enabling verifiable and trustworthy AI-driven insights

Last Outfit - E-commerce Website [\[GitHub\]](#) — Tech Stack: MongoDB, Express, React, Node.js

Aug 2023 – Nov 2023

Designed and developed a full-stack application using OOP principles, implementing authentication, database operations, backend APIs, and testing to improve stability.

Improved product and order data management using **MongoDB** for fast and scalable queries.

Enhanced backend API performance using **Node.js and Express**, reducing latency significantly.

CERTIFICATES

Full-Stack Development ([Certificate](#)), IBM DevOps ([Certificate](#)), Linux for Developers ([Certificate](#)),

AI for Beginners ([Certificate](#)), Machine Learning ([Certificate](#)), Python ([Certificate](#))